



Fig. 5: Motor fitting in the soap disposal pump



PARTS LIST

Semiconductors:

- IC1 - 555 timer
- D1 - IR transmitter
- D2 - IR receiver
- D3-D4 - 1N4004 diode
- T1 - 2N3906 pnp transistor
- T2 - 2N2222 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 4.7-kilo-ohm
- R2 - 33-kilo-ohm
- R3, R5 - 1-kilo-ohm
- R4 - 680-ohm

Capacitor:

- C1 - 10 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- S1 - Pushbutton switch
- M1 - 5V DC motor
- VR1 - 100 kilo-ohm potentiometer

To build this automatic dispenser, we recommend using a self-priming gear motor. If possible, reuse a gear motor pump from an old commercial dispenser whose electronic board may have been damaged.

Constructing and testing

Fig. 3 shows the actual-size, single-side PCB layout for the automatic dispenser using the LM555. The component layout can be seen in Fig. 4. Once you have assembled the circuit on PCB, enclose it in a suitable box. For fitting the motor with the soap disposal mechanism, refer to Fig. 5. Place the switch in front of the cabinet for manual use.

Fig. 6 displays the author's final prototype, excluding the motor and soap disposal pump, which need to be connected externally using wires.

After fixing the motor as shown in Fig. 5 and assembling the components as shown in Fig. 6, interconnect both and install at a proper location based on your choice. **EFY**

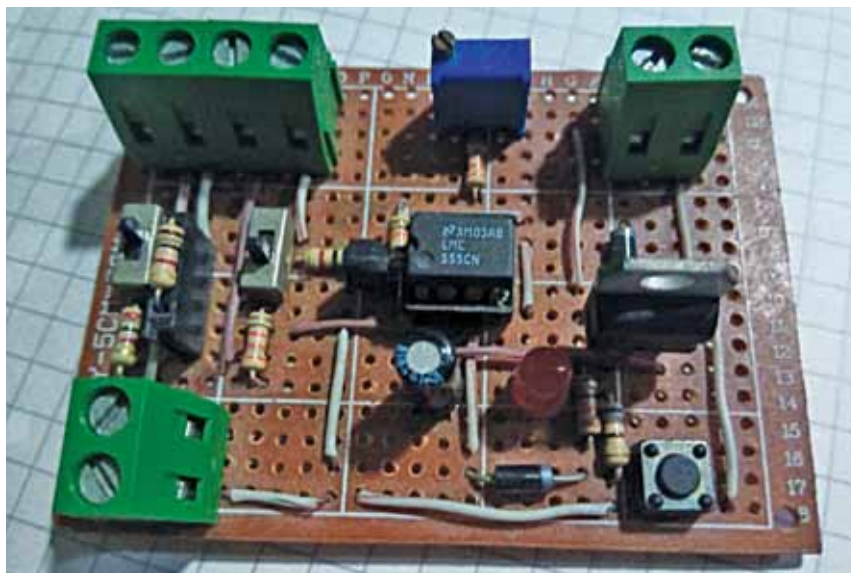


Fig. 6: Author's prototype

sent to transistor T2 to activate the DC motor that dispenses the soap based on the time set by capacitor C1 and potentiometer VR1. It's important to note that this particular monostable mode of the 555 timer sends only one pulse. Even if you don't move your hand (obstructing the line of sight), the only way the automatic dispenser will dispense again is by removing your hand, which turns off the 555 timer. When the line of sight is obstructed again,

the dispenser will dispense soap once more.

The potentiometer in this circuit is used to adjust the dispensing time of the dispenser. We recommend a range of 0.5 to 1.2 seconds. The precise time depends on the tests you conduct and the viscosity of the soap or gel.

In the circuit diagram (Fig. 2), you will notice a push-to-on switch (S1). This switch is used for dispensing without using a sensor.

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